

Cardiovascular System

Physical Diagnosis Lecture

27 June 2016

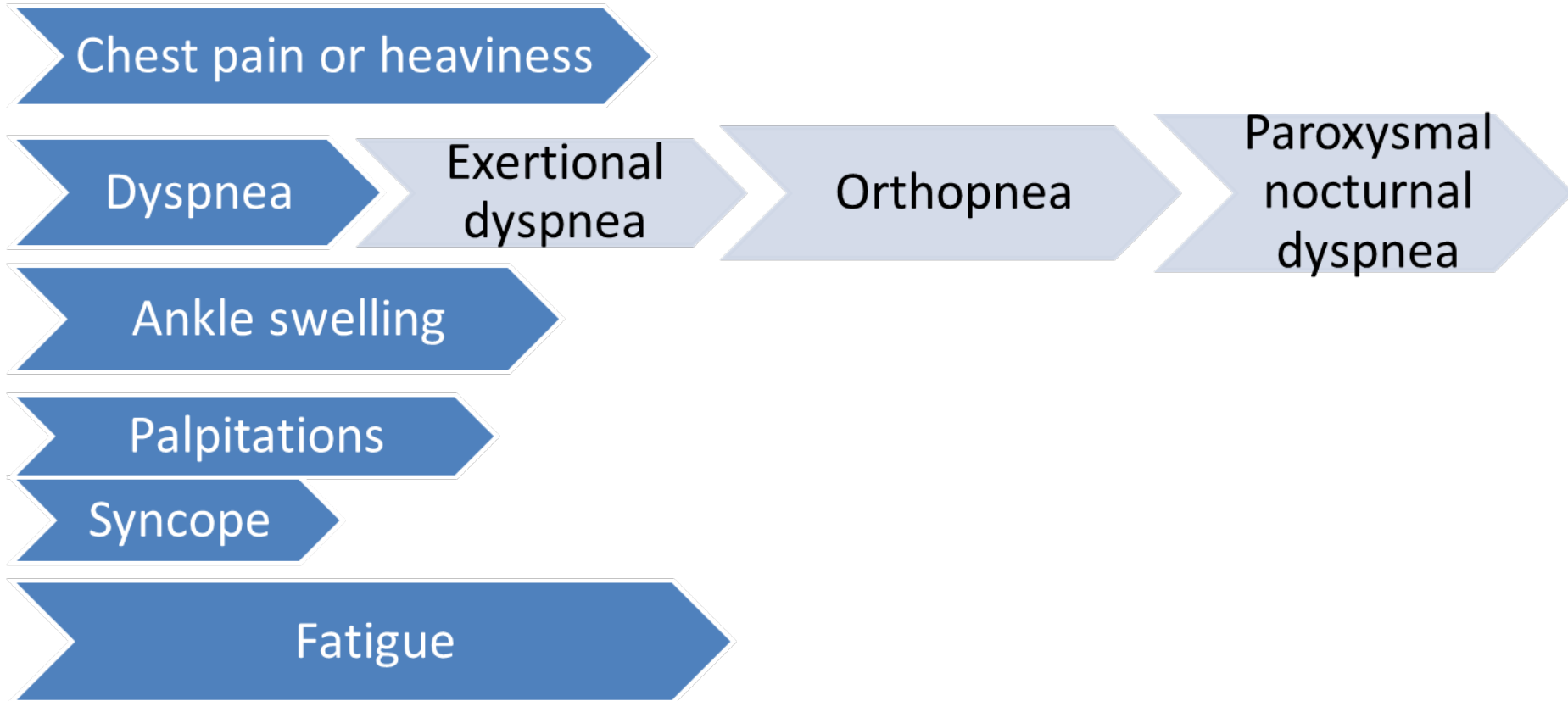
History and examination of the heart and blood vessels

Other parts of body where symptoms and signs of heart disease may appear

1. Cardiovascular History

2. Cardiovascular Examination

Major symptoms



Past history

- History of ischemic heart disease: myocardial infarction
- Rheumatic fever
- Sexually transmitted disease
- Recent dental work, thyroid disease
- Prior medical examination revealing heart disease (e.g. school, insurance)
- Drugs

Social history

- Tobacco and alcohol use or use of other substances
- Occupation

Family history

- Myocardial infarcts
- Cardiomyopathy
- Congenital heart disease
- Mitral valve prolapse

Functional status in established heart disease

Class I-disease present but no symptoms, or angina or dyspnea during unusually intense activity

Class II-angina or dyspnea during ordinary activity

Class III-angina or dyspnea during less than ordinary activity

Class IV-angina or dyspnea at rest

Chest pain

To help determine cause of chest pain:

Duration

Location

Quality

Precipitating & aggravating factors

**Means of relief & accompanying
symptoms**

Radiation

Dyspnea = Shortness of breath

Dyspnea (Greek *dys* 'bad', *pnoia* 'breathing')

Definition: an unexpected awareness of breathing.

Causes:

- Cardiac
- Lung disease or
- other causes (eg. Anxiety).

Orthopnea

Dyspnea that develops when a patient is supine

Causes of orthopnea

- **Cardiac failure**
- **Uncommon causes**
 - Massive ascites
 - Pregnancy
 - Bilateral diaphragmatic paralysis
 - Large pleural effusion
 - Severe pneumonia

Paroxysmal nocturnal dyspnea (PND)

Severe dyspnea that wakes patient from sleep so that he or she is forced to get up gasping for breath.

Ankle swelling

Ankle edema of cardiac origin is usually symmetrical and worst in evenings, with improvement during night

As failure progresses, edema ascends to involve legs, thighs, genitalia and abdomen.

Other common causes of ankle edema:

- Renal diseases; eg. nephrotic syndrome : affects face first
- Liver disease
- Drugs

Palpitations

An unexpected awareness of the heartbeat.

Ask: slow or fast, regular or irregular, and how long they last.

Ask: Sudden Vs gradual onset and offset.

Ask associated features: pain, dyspnea or faintness

Syncope, presyncope and dizziness

Syncope: a transient loss of consciousness

Presyncope: a transient sensation of weakness without loss of consciousness.

Establish whether patient actually loses consciousness & under what circumstances the syncope occurs:

- Postural syncope: on standing for prolonged periods or standing up suddenly
- Micturition syncope: while passing urine
- Tussive syncope: on coughing or
- Vasovagal syncope: with sudden emotional stress

Syncope, presyncope and dizziness

- Find out whether there is any warning, such as dizziness or palpitations, and how long the episodes last.
- Recovery may be spontaneous or the patient may require attention from bystanders.
 - ? abnormal movements (epilepsy , primary syncope).

Fatigue

A common symptom of cardiac failure.

May be associated with a reduced cardiac output and poor blood supply to the skeletal muscles.

Other causes: lack of sleep, anemia, depression.

Risk factors

Risk factors for coronary artery disease (established)

- Previous ischemic heart disease
- Hypercholesterolaemia
- Smoking (recorded as packet years)
- Hypertension (when was it first diagnosed and what treatment, if any, has been instituted?)
- A family history of coronary artery disease
- A history of diabetes mellitus
- Renal failure

Risk factors

Valvular heart disease

- History of dental decay or infection (risk of infective endocarditis).
- Antibiotic prophylaxis before dental (and some surgical) procedures.

Treatment

Medications a patient is taking: often a good clue to diagnosis

Any ill-effects from current or previous medications.

Surgical history

- Previous angioplasty or coronary artery bypass grafting
- If patient is unable to provide a history: a midline sternotomy scar and scars consistent with previous saphenous vein harvesting

Past history

History of previous angina or myocardial infarction

- How was diagnosis of ischaemic heart disease made and in particular what investigations were undertaken.
- Any treatment?

Social history

Whether patient's condition has prevented from working and over what period.

- Patients with severe cardiac failure, for example, may need to make adjustments to their living arrangements
 - so that they are not required to walk up and down stairs at home.

1. Cardiovascular History

**2. Cardiovascular
Examination**

Comprehensive examination.

- First, appropriately expose & position patient properly & get an impression of general appearance.
- Then do detailed examination
 - Begin with hands & pulses & progress smoothly to neck, face & then on to the *precordium*
 - (Latin *pre* 'in front of', and *cor* 'heart').

Positioning patient

Place patient in bed with enough pillows to support him or her at 45 degrees.

- This is usual position in which jugular venous pressure is assessed.

During auscultation, optimal examination requires further positioning of the patient



General appearance

Look at general state of health.

- Does patient appear to be ill?
- Does patient at rest have rapid and labored respiration, suggesting dyspnea?
- Does patient look *cachectic*?
 - Severe loss of weight & muscle wasting (*cardiac cachexia*)).

The hands

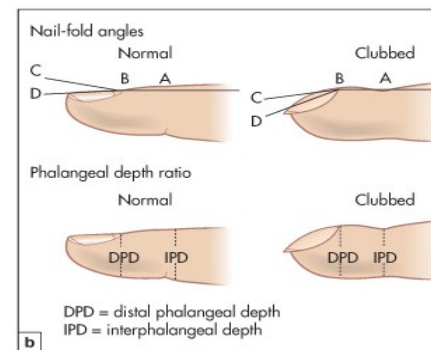
Look first at the nails.

Look for clubbing.

- Inspect fingernails (and toenails) from side to determine if there is loss of angle between nail bed & finger
- the *hyponychial angle* (Figure).



Appearance

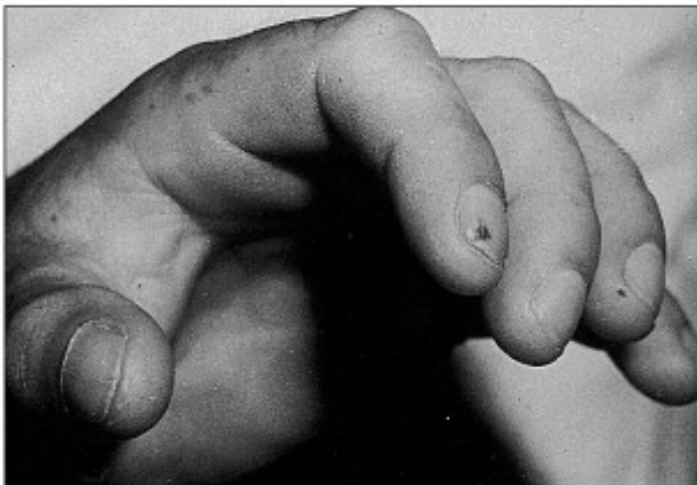


Phalangeal depth ratio

The hands

Nails: look for *splinter hemorrhages* in the nail beds (Figure).

- linear haemorrhages lying parallel to long axis of the nail.



The hands

Osler's nodes

- Red, raised tender nodules on the pulps of the fingers (or toes), or on the thenar or hypothenar eminences.

Janeway lesions (Figure)

- Non-tender erythematous maculopapular lesions containing bacteria, which occur very rarely on palms or pulps of fingers



Arterial pulse

Palpate radial artery at the wrist.

- Just medial to the radius, using forefinger & middle finger pulps of examining hand (Figure)

Observations to be made:

1. Rate of pulse, rhythm

2. Radiofemoral delay

- delay of femoral pulse compared with radial pulse

3. Character & volume of pulse

- better assessed from palpation of brachial or carotid arteries

4. Condition of the vessel wall



Arterial pulse

Rate: Count for 30 seconds or 1 minute to obtain rate per minute.

- Normal resting HR in adults: 60 - 100 beats per minute

Rhythm: regular or irregular.

Radiofemoral delay

- While palpating radial pulse, place fingers of other hand over femoral pulse (below inguinal ligament, one-third of the way up from the pubic tubercle)(Figure).



Arterial pulse

Examine peripheral pulses: radial, brachial, carotid, femoral, popliteal, posterior tibialis, dorsalis pedis



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Figure 3.30 Feeling (a) the posterior tibial artery and (b) the dorsalis pedis artery

Arterial blood pressure (BP)

An essential part of examination of almost any patient.

Usually indirect measurements of systolic & diastolic pressures are obtained with a *sphygmomanometer*

Systolic BP

Diastolic BP

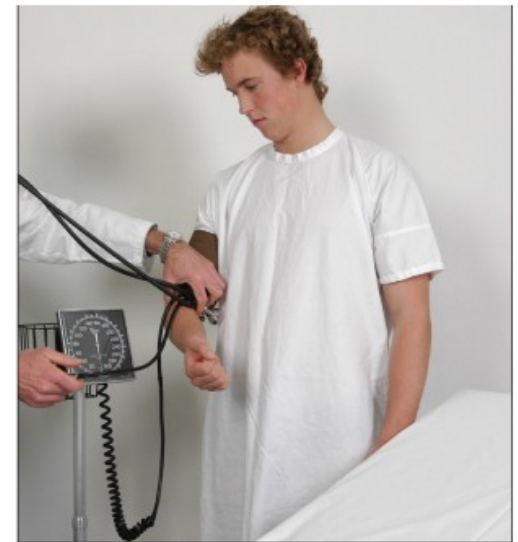
Normal: a systolic reading of < 140 mmHg *and* a diastolic reading of < 90 mmHg.

Postural blood pressure

BP should routinely be taken with the patient both lying down and standing

A fall of more than 15 mmHg in systolic BP or 10 mmHg in diastolic BP on standing is abnormal (*=postural hypotension*).

- It may cause dizziness or not be associated with symptoms.



The face

Inspect sclerae for *jaundice* , *Xanthelasmata* (intracutaneous yellow cholesterol deposits around the eyes) (Figure)

Mitral facies: rosy cheeks with a bluish tinge

Mouth: a *high arched palate* (*congenital heart disease*), teeth, tongue and lips for central cyanosis. Inspect the mucosa for petechiae

Figure 3.12 Xanthelasmata



The neck

- Carotids:
 - provide a great deal of information about the wave form of the aortic pulse, which is affected by many cardiac abnormalities
 - easily accessible, medial to the sternomastoid muscles ([Figure](#))



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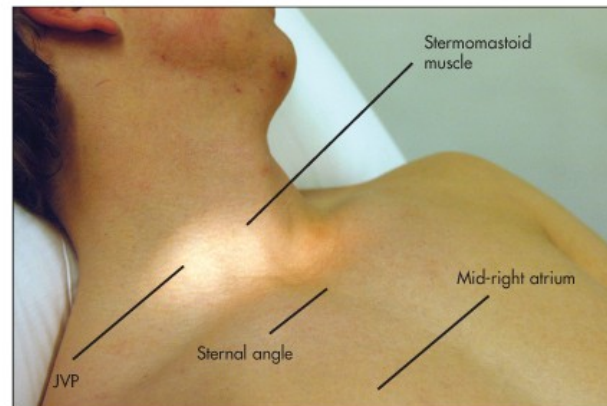
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Figure 3.13 Palpating the carotid pulse

Jugular venous pressure (JVP)- pulsation

- Just as **carotid pulse** tells us about aorta and LV function, **JVP** tells us about right atrial and RV function.
- Positioning of the patient and lighting are important.
- Patient must be lying down at 45 degrees to the horizontal with his or her head on pillows & in good lighting conditions.

Figure 3.14 Assessment of the JVP



JVP

- By convention, sternal angle is taken as the zero point
- Maximum height of pulsations in internal jugular vein, which are visible above sternal angle when patient is at 45 degrees is measured in centimetres.
- In the average person the centre of the right atrium lies 5 cm below this zero point.
 - Normal JVP < 3cm

Precordium Inspection

Inspect first for scars.

- Previous cardiac operations will have left scars on the chest wall.

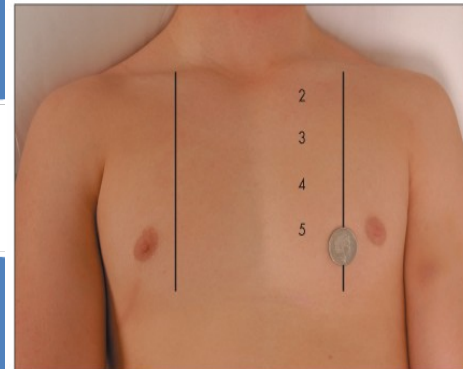
Skeletal abnormalities

- *Pectus excavatum* (funnel chest) or
- *kyphoscoliosis* (a curvature of vertebral column)

Look for the apex beat.

- Normal position is in fifth left intercostal space, 1 cm medial to the midclavicular line (Figure).

Look for other pulsations



Palpation

Apex beat (Figures) vs Point of Maximal Impulse

- The most lateral and inferior point at which the palpating fingers are raised with each systole.
- Palpable in only about 50% of adults.

Count down the number of interspaces.



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Figure 3.18 Feeling for the parasternal impulse

Palpation

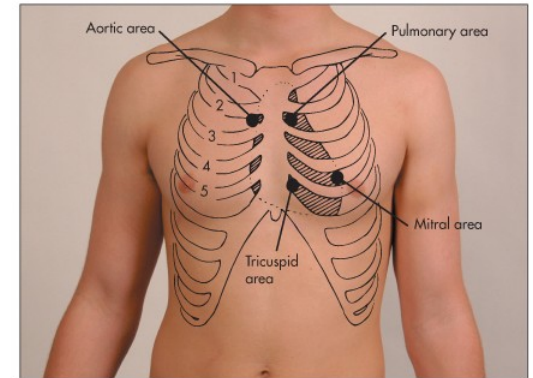
Heart sounds

- Palpation with fingers over pulmonary area may reveal palpable tap of pulmonary valve closure (*palpable P2*) (Figure).

Thrill: palpable murmurs (Fig)

Heave

- A *parasternal impulse* felt when heel of hand is rested just to left of sternum with fingers lifted slightly off the chest (Figure).



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Figure 3.20 The areas best for auscultation do not exactly correlate with the anatomical location of the valves.

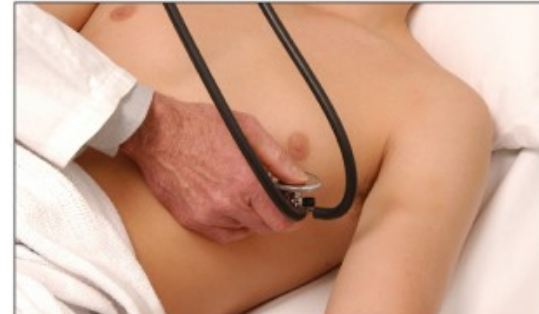


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Figure 3.19 Palpating the base of the heart

Auscultation

- Stethoscope is needed.
- Begins in mitral area with bell of stethoscope (Figures).
- Next, listen in mitral area with diaphragm of the stethoscope ([Figure](#))



Auscultation

- Then place stethoscope in tricuspid area (fifth left intercostal space) and listen.
- Next, left sternal edge to pulmonary (second left intercostal space) and
- Aortic (second right intercostal space) area ([Figure s](#)), listening carefully with diaphragm.



Auscultation

- Heart sounds (S1 and S2)
- Added sounds (murmurs, gallop, friction rub)

Other parts of the body

Back

- percussion and auscultation of the *lung bases*.

Abdomen

- liver, spleen, ascites, hepatojugular reflux, pulsations

Lower limbs

- edema, cyanosis, clubbing



Thank You!